

Rank-Before-Merge isolates the ranking stage

Same input, budget, L2 selector, and native merger

Shared visual input



One image, two controlled paths

ViT units (N)
merger input

a RBM (ours): rank before the native merger

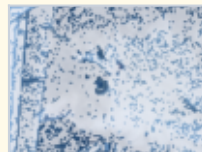
1 **MEASURED L2 RANK**
on merger-input 2x2 units



2 **KEEP TOP- κN**
 $\kappa = 25\%$ of N units



3 **NATIVE MERGER**
applied only to survivors



b Post-merger control: merge first, then rank

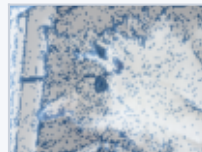
1 **NATIVE MERGER**
applied to all N units



2 **MEASURED L2 RANK**
on merged outputs



3 **KEEP TOP- κN**
 $\kappa = 25\%$ of merged tokens



■ solid border: kept

□ faded: discarded

■ blocky tile: merged token

Controlled comparison: same image, model path, κ , L2 selector, and native merger; only the score tap point changes.

Qwen3-VL-8B | measured merger features | 25% kept

color: per-arm min-max | selection: raw measured L2 (NPZ keep)

No attention or query signal is depicted.